

Description of China Household Finance Survey Data

Variable Rules

2015

I. Data Cleaning

1. Cleaning process

After the data is collected, the center will conduct the preliminary processing on all data, mainly including: deleting invalid samples due to serious groundless answer and cheating by interviewers, deleting invalid variables, deleting sensitive data, correcting duplicate sample numbers caused by humans, and correcting interviewers' active reports that are man-made misuse; merging, tracking new visiting data, splitting household and personal data, splitting multiple choices; adding labels, adding questionnaire types; cleaning up notes and other options, confirming the numerical value interpolation and related rules.

After preliminary processing, the data generates a usable version. If the extreme value or abnormal value is found again during the data utilization, the center will check the secondary recording for confirmation. If there is no recording, the extreme value and abnormal value are processed by creating a model to update the data version and inform the users.

2. Data storage

After completing the data cleaning, the data of China Household Finance Survey can be stored in the following 3 data sets.

1. The data set file name contains “**hh**”, which represents the data of the household part within the questionnaire, such as: assets and liabilities, household expenditures and income, financial knowledge, grassroots governance and subjective evaluation;
2. The data set file name contains “**ind**”, which represents the data of the personal part within the

questionnaire, such as: demographic characteristics (partial), personal work and income information, insurance and security, family member education, etc.

3. The data set file name contains “**master**”, which represents the regional data, specifically including the province information of the investigation and the weight of each investigation data.

In addition, the original data usage format corresponds to the dta format (which can be opened with stata13.0 [best version] or above).

II. Variable Naming Rules

The variable name of the data of China Household Finance Survey questionnaire consists of the initial letter and the last four digits. The different initials correspond to different parts of the questionnaire (e.g. b-agriculture/business; d-financial assets), some special identification variables (ID variables, urban and rural areas, provinces, etc.) are renamed according to their meaning for use. Some variables are followed by the suffix letter it after the original variable name, indicating the inquiry of the range of values for the previous question. Each question in the questionnaire is given the corresponding variable name. The variables that need special attention and their naming rules are described as follows:

1. ID variables

ID variables contain hhid and pline.

The hhid identifies the household variable. Each hhid represents a household, and can exclusively identify a household. The pline is to identify the variables of family members in each family. Each pline represents a family member; hhid and pline can uniquely identify individuals, and the pline of the same family members of the same family remain the same at different times.

2. Weight (swgt)

Under our sampling design plan, the number of households represented by each household is different because the probability of each household being drawn is different. When inferring the

total, it is necessary to adjust the weights to truly and accurately reflect the number of households represented by each household sample to obtain a correct inference of the total. All calculation results of the China Household Finance Survey have been adjusted regarding the sampling weights.

There is a weight in the master data set: swgt.

The calculation method of the sampling weight is as follow: according to the sampling of each stage, the probability P_1 to be drawn of the city and the county under investigation, and the probability P_2 to be drawn of the community (village) under investigation in its district or county and the probability P_3 to be drawn of the sample under investigation in its community (village) are calculated respectively. Three-stage sampling weights are calculated respectively $W_1=1/P_1$, $W_2=1/P_2$, $W_3=1/P_3$, and finally the sampling weight of the sample is $W=W_1 \times W_2 \times W_3$.

That is swgt = the reciprocal of the probability that this community is drawn.

3. Other options in the variables

In data variable single-choice or multiple-choice questions, the last option in general is “Others (please specify it)”. If the respondent chooses this option, there will be one more fill-in-the-blank question. In the 2015 China Household Finance Survey data, the newly generated variable naming format is: the original variable followed by the suffix –ex1, for example: b1001c_ex1 indicates that the respondent should be classified into other related agricultural operating household.

4. Cyclic questions

For all cyclic questions, the naming convention is to add the suffix _# after the original variable name; # represents the # cyclic times. For example: c2003_1 indicates the floor area of the first set of house; thus c2003_2 indicates the floor area of the second set of house.

Table 1 Variables List of Cyclic Questions

Varname	Cycle index	Variable Information
b3001-b3054it	2	varname_1 agricultural loan
		varname_2 commercial loans
c2000-c2000ait	3	varname_#The information of No.# house
c2003-c2022	3	property
		varname_#The information of No.# house

property

Note: The questionnaire also contains other cycles generated due to loading, such as c5004hf, which are not listed here.

5. Multiple-choice questions

For multiple-choice questions, the processing principle is to convert each option to a dummy variable with values 0 and 1. Multiple-choice questions are divided into two categories: acyclic multiple-choice questions and cyclic multiple-choice questions.

(1) Acyclic multiple-choice questions

The naming rule for acyclic multiple-choice questions is to add the suffix `_*_mc` after the original variable name; * represents the * option. For example: b1004_1_mc indicates whether the first option "food crop" of b1004 is ticked; 0 means not selected, 1 means selected.

Table 2 Variable List of Acyclic Multiple-choice Questions

Varname	Number of choices	Varname	Number of choices	Varname	Number of choices
a4007d	6	c1000ba	15	d7113	6
a4007f	6	c1013	9	e1007aa	8
a4007g	5	c3004a	10	e1020aa	8
a4007i	5	c3017d	10	e2002ab	6
a4007k	9	c5001a	6	e2004a	8
a4007q	5	c5004he	13	e2006	6
a4007r	5	c5007e	7	e2013a	6
a4007s	6	c5014a	9	e2013b	5
a4012	8	c5015a	7	e2019a	14
a4013	8	c5021	6	e2020b	4
a4026	11	c5024	6	e3003	9
a4028	6	c5027	4	e3005b	9
a4030	11	c5033	8	e3016	7
b1001c	6	c5033a	8	e3020	5
b1004	6	c5034	9	e3021a	8
b1004b	6	c5040	5	g1009b	13
b1004c	12	c7071	14	g2002	10
b1004g	17	c8001	17	g2004	7
b1009	6	c8004	7	h1002	10
b1014a	4	d2107	7	h1004	8
b1017	14	d2108	9	h2001	12
b1102	5	d2118	10	h2003	8

b1104	11	d2118a	9	h3014	5
b1105	7	d2118b	7	h3024	3
b1106	11	d2126	7	h3045	7
b1107b	9	d2130	10	j2001	6
b1108	6	d3102aa	10	j2004	5
b1116	6	d3105a	10	j2011	4
b2000c	8	d3111b	9	k2119	9
b2008	6	d4101a	5	a3049a	8
b2078	13	d5102aa	11	a3052	15
b2083c	7	d5104	5	f2001a	12
b3055a	4	d7106e	11	f6001	4
b3061	5	d7106g	8		

Table 3 Processing Instruction of Acyclic Multiple-Choice Questions

varname_*_mc	Variable Information
0	The * option has not been selected
1	The * option has been selected

(2) Cyclic multiple-choice questions

The naming rule for cyclic multiple-choice questions is to add the suffix `_*_#_mc` after the original variable name; * represents the * option, while # represents # cycle. For example, `b3002a_2_1_mc` indicates whether the second option "Estimated that loan application will not be approved" is ticked in the first cycle; 0 means not selected, 1 means selected.

Table 4 Variable List of Cyclic Multiple-choice Questions

Varname	Number of Choices	Cycle Index	Varname	Number of Choices	Cycle Index
b3002a	9	2	b1001d	3	3
b3003	9	2	c2005a	3	3
b3012	10	2	c2010d	10	3
b3014a	8	2	c2011b	3	3
b3021c	8	2	c2000ai	10	6
b3021d	7	2	c2000ak	3	6
b3027a	8	2	c2000ap	3	6

Table 5 Processing Instruction of Cyclic Multiple-Choice Questions

varname_*_#_mc	Variable Information
0	The * option has not been selected in the # cycle.
1	The * option has been selected in the # cycle.

6. Interpolation variable

In order to solve the problem of data missing, we made the interpolation processing for questions with “it”. For the interpolated variable, the naming rule is the original variable name plus the suffix _imp.

Table 6 Variable List of Interpolated Questions

Varname				
b1004h_#_imp	b3030c_#_imp	a3037_imp	c2027d_imp	d7112_imp
b1004_#_imp	b3031a_#_imp	f1005_imp	c2032_imp	d8104_imp
b2036_#_imp	b3044_#_imp	f1010_imp	c2041_imp	d8106_imp
b2037_#_imp	b3044a_#_imp	f1031_imp	c3002_imp	d9100aa_imp
b3002b_#_imp	b3051_#_imp	f1036_imp	c3002a_imp	d9100b_imp
b3005_#_imp	b3054_#_imp	f2006_imp	c3016_imp	d9110a_imp
b3005a_#_imp	c2000a_#_imp	f4005_imp	c3003_imp	d9110b_imp
b3008_#_imp	c2000f_#_imp	f4008_imp	c3017ca_imp	d9108_imp
b3008f_#_imp	c2013_#_imp	f4011_imp	c3019a_imp	e1006_imp
b3018_#_imp	c2016_#_imp	b1005b_imp	c3019c_imp	e1006a_imp
b3020_#_imp	d4103_#_imp	b1012_imp	c3019e_imp	e1007a_imp
b3022a_#_imp	b1004h_#_imp	b1013a_imp	c3024_imp	e1016_imp
b3022b_#_imp	b1004_#_imp	b1013c_imp	c3025_imp	e1022_imp
b3030c_#_imp	c2000a_#_imp	b1013e_imp	c7009_imp	e1021_imp
b3031a_#_imp	c2000f_#_imp	b1014_imp	c7022_imp	e1034_imp
b3044_#_imp	c2013_#_imp	b1015a_imp	c7023_imp	e1027_imp
b3044a_#_imp	c2016_#_imp	b2021_imp	c7077_imp	e1046_imp
b3051_#_imp	d4103_#_imp	b2022_imp	c7078_imp	e3003c_imp
b3054_#_imp	b1004h_#_imp	b2040_imp	c7052b_imp	e3005c_imp
c2000a_#_imp	b1004_#_imp	b2041_imp	c8007_imp	e3019_imp
c2000f_#_imp	c2000a_#_imp	b2046_imp	d1105_imp	e4003_imp
c2013_#_imp	c2000f_#_imp	b2050_imp	d2104_imp	e4004_imp
c2016_#_imp	d4103_#_imp	b2052_imp	d2106_imp	e4005_imp
d4103_#_imp	b1004h_#_imp	b2055_imp	d3103_imp	e4006_imp
b1004h_#_imp	b1004_#_imp	b2059_imp	d3109_imp	e4018_imp
b1004_#_imp	c2000a_#_imp	b2063_imp	d3110_imp	g1014_imp
b2036_#_imp	c2000f_#_imp	b2080_imp	d3116_imp	g1015_imp
b2037_#_imp	d4103_#_imp	b2084_imp	d3117_imp	g1017_imp
b3002b_#_imp	b1004h_#_imp	b2085_imp	d4111_imp	g1018_imp
b3005_#_imp	b1004_#_imp	b2093_imp	d5107_imp	g1019_imp
b3005a_#_imp	c2000a_#_imp	b2103_imp	d5108_imp	g1019a_imp
b3008_#_imp	c2000f_#_imp	b2110_imp	d5109_imp	h2004_imp
b3008f_#_imp	a3020_imp	c2002d_imp	d6100a_imp	k1101_imp
b3018_#_imp	a3022_imp	c2023d_imp	d6116_imp	k2102b_imp

b3020_#_imp	a3023_imp	c2027a_imp	d7106a_imp	k2114_imp
b3022a_#_imp	a3024_imp	c2027b_imp	d7106h_imp	
b3022b_#_imp	a3036_imp	c2027c_imp	d7110_imp	

7. Truncation

In accordance with the provisions of the statistical law, in order to protect the privacy of respondents, we have truncated the extreme values of income and assets. Replace the sample whose income or assets exceed the specific value with the specified value, and at the same time give a truncated dummy variable. The naming rule is to prefix the variable name with censor_, i.e. censor_varname. For example, we replaced the annual income (hh_income) of the 2011 China Household Finance Survey data with more than 3 million in 3 million, and also gave a dummy variable censor_hh_income, 0 means no truncation; 1 means truncation.

Table 7 Instruction to Truncation

censor_varname	Variable Information
0	Unprocessed
1	Processed

8. Description of Other Special Variables

Table 7 Description of Special Variables

Name of Variable	Value	Meaning of Value	Supplementary Note
hhead	0	Non-householder	
	1	Householder	
respond	0	Non-respondent	“remark” variable: Respondents of unknown type are included in the remarks of "remark variable"
	1	Respondent	
rural	0	Urban	
	1	Rural	
istrack	0	New interview	
	1	Follow-up interview	
aorb	a	Sub-questionnaire a	
	b	Sub-questionnaire b	
region	1	East China	Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, Hainan

2	Central China	Shanxi, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, Hunan
3	West China	Inner Mongolia, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, Qinghai, Ningxia

9. Description of Assets, Debts, Income and Consumption Variables

Total household assets (asset) include non-financial assets (nfasset) and financial assets (fasset). Non-financial assets include agricultural operating assets, industrial and commercial operating assets, land assets, real estate, vehicle assets, and other non-financial assets. Financial assets include social security account balances, cash, deposits, stocks, funds, bonds, derivatives, wealth management, foreign currency assets, gold, other financial assets, and lending.

Total household debts (debt) include agricultural debt, business debt, real estate debt, vehicle debt, other non-financial assets debt, stock debt, other financial assets debt, education debt, medical debt and other debts.

Total income (total_income) of a household includes wage income, agricultural operating income, industrial and commercial operating income, transfer income, and investment income.

Total household consumption expenditures (total_consump) include food expenditure, clothing expenditure, housing expenditure, living goods and services expenditure, education and entertainment expenditure, transportation and communication expenditure, medical care expenditure and other expenditures.

10. Description of Missing Values

The actual interview is affected by subjective and objective factors. Due to the subjective knowledge and attitude of the respondent, the wrong understanding or wrong answer of the interviewer and other reasons, a small amount of missing values will inevitably occur. These missing values are defined as follows:

1. The data value is ".d"

The respondent does not know how to answer. The interviewer chooses "unknown" (this option is not included in the actual questionnaire). Missing value caused by unknown.

2. The data value is ".r"

The respondent refuses to answer the question, and the interviewer chooses "refuse to answer"

(this option is not included in the actual questionnaire). Missing value caused by refusal to answer.

III. Description of Respondent Batches and Sub-questionnaires

1. Description of Respondent Batches

As students were divided in two batches, the 2015 household financial questionnaire was revised when the second batch of students join the survey. The specific adjustment of variables is listed as follows:

Table 8 Description of Variables in Questionnaire of Each Batch

S/N	Type	Questionnaire of the first batch	Questionnaire of the second batch
1	Delete variable	a3037, a3037it	Delete a3037 and a3037it
2	Delete variable	b1014aa, b1014ab, b1014ac, b1014ad	Delete b1014aa, b1014ab, b1014ac and b1014ad
3	Delete variable	b1009	Delete b1009
4	Delete variable	c1000ac	Delete c1000ac
5	Delete variable	c5003a, c5004	Delete c5003a and c5004
6	Delete variable	c5004h	Delete c5004h
7	Delete variable	c5004hd, c5004he, c5004hf, c5004hf_01, c5004hg	Delete c5004hd, c5004he, c5004hf, c5004hf_01 and c5004hg
8	Delete variable	c5004hm, c5004hn, c5004hp	Delete c5004hm, c5004hn and c5004hp
9	Delete variable	c5040, c5043, c5044, c5045, c5046, c5047, c5048, c5049	Delete c5040, c5043, c5044, c5045, c5046, c5047, c5048 and c5049
10	Delete variable	c7008a	Delete c7008a
11	Delete variable	b1115, b1116	Delete b1115 and b1116
12	Add variable	No a4007a-a4007t Jinhui questionnaire	Add a4007a-a4007t Jinhui questionnaire

13	Change order	a2018b, a2019a, a2019b	Change order into: a1919b, a2019a, a2018b
14	Change order	a2025b, a2025ba, a2025bb	Change order into: a2025ba, a2025bb, a2025b
15	Adjust options	b3012: options 1-8, 8 as "Others"	b3012: options 1-10, 10 as "Others", add: 8 as "Homestead" and 9 as "Collective forestry property right"
16	Adjust options	b2001a: options 1-5, 5 as "Others"	b2001a: options 1-6, 6 as "Others", add: 5 as "Inherit family property"
17	Adjust options	B3061: options 1-4, 3 as "Margin and 4 as "Others"	b3061: options 1-5, 3 as "Warranty", 4 as "Advance money on security" and 5 as "Others"
18	Adjust options	b3060	b3060: add option 3 "Not willing/not necessary to loan from the bank"
19	Adjust options	g1009b: options 1-10, 10 as "Others"	g1009b: options 1-13, 12 as "Others", add: 10 "Make and receive phone calls", 11 "Send and receive SMS", 13 "Above functions are not available"
20	Adjust options	h3055 "Are you willing to receive quarterly follow-up phone call?"	h3055 "Accepted frequency to receive follow-up phone call: monthly, quarterly, semi-annual, annual, not willing"
21	Modify question type	c1000af Multiple choices	c1000af Fill in the blank
22	Reset questioning condition	Enter question a3015 when a3014a=3	Do not ask question a3015 when a3014a=3
23	Reset questioning condition	Do not ask question a3018a when a3014a=3	Enter question a3018a when a3014a=3
24	Reset questioning condition	Question b1001a for both urban and rural samples	Question b1001a for only rural samples
25	Reset questioning condition	No limit conditions for questions c8006-c8007it	Only ask question c8006-c8007it when c8001≠18 or c8004≠7
26	Reset questioning condition	No limit conditions for questions d3101-d3117it	Do not ask questions d3101-d3117it when a4007=3 or 5
27	Reset questioning condition	No limit conditions for questions d5102-d5109it	Do not ask questions d5102-d5109it when a4007=4 or 5

28	Reset questioning condition	No limit conditions for questions a1001-a1009	Do not ask questions a1001-a1009 for Ali samples
29	Reset questioning condition	a2032-a2041	Question a2032-a2041 only for respondents aged 60 and below
30	Reset questioning condition	a2042	Do not ask question a2042 when a2035b=2

2. Description of Sub-questionnaires

Some variables in the 2015 household financial questionnaire are divided into sub-questionnaires a and b, which are specifically listed as follows:

Table 9 Description of Division of Variables in Sub-questionnaires a/b

Sub-questionnaires a only	Sub-questionnaires b only
a4012, a4014, a4016, a4017, a4020, a4022, a4024, a4026, a4026a, a4026b, a4030, a4032, a4034, c1000ac, c1000ab, c1000ac, c1000ad, c1000ae, c1000af, c1000ag, c1000ah, c1000ai, c1000ba, c1000bb, c1000bc, c4001, c4002, c4004, c4005, c4006, c4007, c4007a, c4008, c4008b, c4008c	a4013, a4015, a4015a, a4018, a4019, a4021, a4023, a4025, a4027, a4027a, a4027b, a4027c, a4031, a4033, a4035, b2056, c2000ae, c2000af, c2000ag, c2000ah, c2000ai, c2000aj, c2000ak, c2000al, c2005a, c2010a, c2010b, c2010c, c2010ca, c2010d, c2010da, c2010e, c3018, c3019, c3019aa, c3019a, c3019ait, c3019b, c3019c, c3019cit, c3019d, c3019e, c3019eit, c3024, c3024it, c3025it, c3025it